

Chapter 11

Quantum Spin Chains

Bibliography

Index

- algebra, 26, 114
 - C^* , 27, 114
 - Banach, 26
- algebraic quantum theory, 25
- anti-lattice, 73
- anti-linear, 61
- anti-unitary, 61
- Artin-Schreier theory, 63
- automorphism group, 16

- Banach algebra, 26, 129
- basis
 - Hermitian operators, 43
 - nice error, 45
 - shift-and-multiply, 45
 - unitary operator, 43
- Bell state, 45
- Bell telephone, 20
- bicommutant, 28
- biquadratic form, 63
- black box, 18
- Bloch sphere, 10
- boost, 51

- C^* -algebra, 27
- canonical commutation relations, 61
- Cauchy-Schwarz inequality, 131
- center, 16, 28
- Cesaro mean, 94
- Chernoff bound, 140
- χ^2 divergence, 137
- Choi matrix, 18, 33
- classical mechanics, 22
- closed system, 15
- coherent state, 47
- commutant
 - of Kraus operators, 114
- complementary slackness, 70
- complete boundedness, 31
- complete contractivity, 31
- complete positivity, 17, 31
 - conditional, 122
- complex numbers, 24
- concatenation, 42
- concave
 - strictly, 135
- concavity, 79
- condition number, 149
 - Jordan, 153
- conditional complete positivity, 122
- conditional expectation, 30, 115
- cone, 143
 - convex, 143
 - dual, 143
 - proper, 143
- congruence map, 64
- conic program, 69
- convergence
 - strong, 119
 - uniform, 119
 - weak, 120
- convex
 - cone, 143
 - cone, 54
- convexity, 79
 - states, 9
- Csiszar f-divergence, 136

- decomposable map, 64
- density matrix, 9
- determinant, 96
 - channel with negative determinant, 97
- diagonalizable, 78

- dilation space, 38
- diophantine approximation, 95
- doubly stochastic, 133, 135
- doubly-stochastic, 34
 - map, 48
- dual, 8
 - cone, 143
 - program, 70
- dual norm, 132
- duality
 - state-channel, 33, 34
 - strong, 70
 - weak, 70
- effect operator, 11, 19
- Einstein locality, 10, 20
- entanglement witness, 57, 64
- entropy
 - increasing maps, 135
- environment, 18
- expectation value, 8
- f-divergence, 136
- factor
 - von Neumann algebra, 28
- faithful, 27
- filtering operation, 19, 48
- fixed point
 - full rank, 112, 115
 - positive, 110
 - pure, 112
 - support, 111
- flip, 53
- flip operator, 14
- frame, 45
 - tight, 46
- functional, 27
- functional calculus, 78
- Gell-Mann matrix, 43
- generalized coherent state, 47
- generator
 - of a semigroup, 120
- GHZ state, 45
- GNS construction, 28
- group
 - Heisenberg-Weyl, 44
- Hölder's inequality, 66
- Hamiltonian, 16
- Heisenberg picture, 15
- Heisenberg-Weyl group, 44
- Hellinger divergence, 137
- Hermitian operator basis, 43
- Hermiticity preserving map, 34
- Hilbert Schmidt
 - class, 9
 - norm, 9
- Hilbert's 17th problem, 63
- Hilbert's projective metric, 143
- Hilbert-Schmidt, 42
 - scalar product, 9, 42
- inequality
 - Araki-Lieb-Thirring, 131
 - Cauchy-Schwarz, 131
 - Lidskii, 130
 - Schwarz, 75
 - variance, 131
- infinitesimal generator, 16
- informationally complete POVM, 46
- instrument, 18, 19
 - Lüders, 19
- interaction picture, 15
- irreducible
 - non-negative matrix, 99
 - positive map, 99
- Jamiolkowski state, 18, 33
- joint convexity, 84
- joint measurability, 11
- Jordan block, 92
- Jordan condition number, 153
- Jordan decomposition, 92
- Klein four-group, 45
- Kossakowski matrix, 125
- Kraus
 - representation, 36
- Kullback Leibler, 137
- Ky Fan dominance, 131
- l_1 norm, 137

- Löwner's partial order, 73
- Lüders instrument, 19
- Laplace transform, 95
- lattice, 73
- Lemma
 - Stein, 141
- Lindblad operator, 125
- linearity, 17
- Liouville equation, 22
- locality, 20
- Lorentz transformation, 50
- majorization, 132, 133
 - weak, 133
- marginal, 12
- Markovian, 18
 - continuous time evolution, 119
- matrix norm, 129
- matrix unit, 42
- maximally entangled state, 14
- maximally mixed state, 9
- measurement, 7
 - generalized, 11
 - Lüders, 19
 - POVM, 11, 19
 - von Neumann, 11, 41
- measurement problem, 8, 15
- memory-less, 18
- metric
 - Hilbert's projective metric, 143
- minimal dilation, 38
- Minkowski space, 50, 74
- mixed state analyzer, 10, 21
- mixture, 9
- moment inequality, 131
- multiplicative domain, 78
- n-positive map, 57
- Neyman-Pearson, 139
- nice error basis, 45
- no-cloning theorem, 22
- no-signaling, 20
- non-linearity, 21
- norm
 - C^* -norm, 27
 - L_p , 130
 - l_p , 130
 - dual, 132
 - Frobenius norm, 130
 - Hilbert-Schmidt norm, 130
 - Ky Fan, 129
 - matrix norm, 129
 - operator, 130
 - Schatten, 129
 - sub-multiplicative, 129
 - trace norm, 130
 - unitarily invariant, 129
- normal operator, 26, 76
 - quasinormal, 76
 - subnormal, 76
- normal state, 28
- normed algebra, 26
- observable, 8
- one-parameter group, 16
- open system, 15
- operator basis, 43
- operator convexity, 79
- operator monotonicity, 79
- operator system, 31
- orthostochastic, 133
- overcomplete, 45
- partial order
 - Löwner, 73
- partial trace, 12
- partial transpose, 14, 53, 58
- Pauli matrix, 43
- peripheral spectrum, 93, 103
- Perron-Frobenius theory, 105
- pinching, 30
- Poincaré sphere, 10
- polar decomposition, 51
- polarization identity, 31
- positive
 - n -positive, 54
 - element of an algebra, 26
 - functional, 27
 - map, 53
 - polynomial, 63
- positive map
 - irreducible, 99

- positivity, 30, 53
- POVM, 11, 40
 - informationally complete, 46
 - Neumark's theorem, 40
 - SIC, 45, 47
- PPT, 58
- preparation, 7
- primitive, 99
- primitive map, 105
- probabilistic operation, 18
- projection
 - sub-harmonic, 112
- projective diameter, 148
- pure, 27
- purity, 9
- quantum channel, 18
- quasinormal, 76
- qubits, 10
- recurrence, 94
- reduced density operator, 12
- reduction criterion, 59
- relative entropy
 - α entropy, 137
 - classical, 137
 - Renyi, 137
- representation
 - of C^* -algebra, 27
- resolution of the identity, 11
- resolvent, 95
- Schlieder property, 21
- Schmidt coefficients, 13
- Schmidt decomposition, 13
- Schmidt number, 57
- Schrödinger picture, 15
- Schwarz inequality, 75, 114
- Schwarz map, 77, 113
- selfadjoint, 26
- semidefinite program, 69
- semigroup, 119
 - dynamical, 119
 - one-parameter, 119
- SIC POVM, 45, 47
- singular value decomposition, 129
- spectral radius, 27, 91
- spectrum, 26, 91
 - peripheral, 93, 103
- spinor map, 50
- state, 8, 9
 - maximally entangled, 14
 - Bell, 45
 - GHZ, 45
 - maximally mixed, 9
 - normal, 9, 28
 - on C^* -algebra, 27
 - singular, 9
 - stationary, 110
- state vector, 9
- stationary state, 110
- stationary subspace, 111
- steering, 14
- Stinespring representation, 38
- stochastic map, 19
- Stokes parameter, 10
- Stone's theorem, 16
- subnormal, 76
- subspace
 - stationary, 115
- subsystem
 - invariant, 115
- sum of squares, 63
- swap operator, 14
- symmetric gauge function, 129
- teleportation
 - classical, 22
 - entanglement assisted, 35
- theorem
 - Ando-Lieb, 86
 - bipolar, 143
 - Birkhoff, 133
 - Brouwer, 118
 - Chernoff bound, 140
 - Dirichlet, 95
 - Doebelin, 147
 - Hahn-Banach, 32
 - Hiai-Petz, 141
 - Hille-Yoshida, 121
 - Neumark, 40
 - Neyman-Pearson, 139

- no-cloning, 22
- Reeh-Schlieder, 14
- Schauder, 118
- Stein's lemma, 141
- Stinespring, 38
- Stone, 16
- Wigner, 16, 97
- time, 15
- time reversal, 60
- trace non-increasing, 19
- trace preserving, 17
- transfer matrix, 33, 42
- transposition, 53, 58
 - determinant, 96
- unistochastic, 133
- unitary conjugation, 96
- unitary operator basis, 43
- vacuum, 28
- variance inequality, 131
- von Neumann
 - measurement, 11
- von Neumann algebra, 28
- wave function, 9
- Weyl system, 44
- Wielandt's inequality, 107
- Wigner function, 60
- witness, 57
 - decomposable, 58
- z-transform, 95
- Zwanzig projection, 24